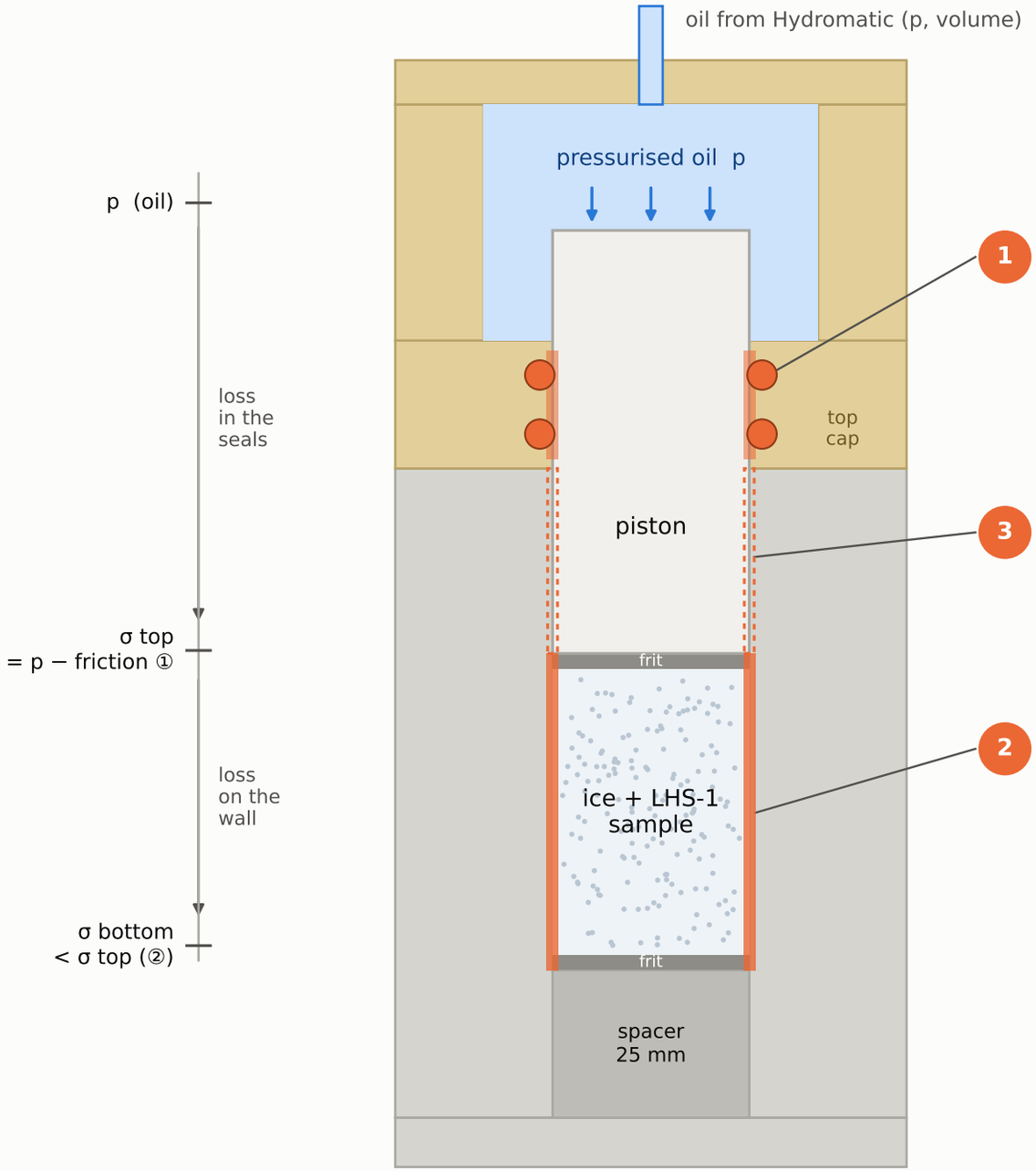
Uniaxial compaction vessel — layout and stress losses

Schematic section, not to scale. Bore 25.4 mm, well depth 96 mm, frits 1.65 mm, bottom spacer 25 mm.



1 **Piston O-ring seals — main friction**

Measured in a test without sample:
- ~1330 kPa at breakaway
- ~210–420 kPa while sliding

(see Chapter 3)

3 **Piston tip – wall (possible)**

Only with frost, dirt or misalignment.

2 **Sample – chamber wall**

Reduced by vacuum grease, not removed. Not measured.
Carries part of the load: stress decreases from top to bottom of the sample.

Measured quantities

Oil pressure p and oil volume V (controller)
Piston displacement: external LVDT, and oil volume / 0.507 cc/mm
Sample temperature: RTD below the sample